Your Name

Solutions

Your Signature

Student ID #

- Please turn off all cell phones.
- This exam is closed book. You may use one two-sided sheet of handwritten notes. You may not use any calculators.
- Please place a box around your answer to each question.
- In order to receive credit, you must *show all of your work*. If you do not indicate the way in which you solved a problem, you may get little or no credit for it, even if your answer is correct.
- If you need space to work on a given problem, use the backs of the pages and indicate that you have done so.
- Cheating will result in a zero and will be reported to the Dean's Academic Conduct Committee.
- This exam has 5 pages, plus this cover sheet. Please make sure that your exam is complete.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total	50	

1. (10 points) Felix is walking on the surface $z = f(x, y) = \frac{1}{4}x^2 + \frac{1}{3}y^3 - \frac{1}{2}xy^2$, where x, y and z are in miles. Label the positive y-direction as NORTH and the positive x-direction as EAST.

4 (a) Felix's x and y coordinates are given by $x = x(t)$ and $y = y(t)$ where t is in hours since he started walking. We are told that $x(1) = 2$, $y(1) = 1$, $x'(1) = 3$ and $y'(1) = -4$. At $t = 1$ hour, find the rate of change of Felix's height with respect to time. (Give units for your answer.)

$$\nabla f = \left(\tfrac{1}{2}x - \tfrac{1}{2}y^2,\ y^2 - xy\right) \qquad \nabla f(2,1) = \left(\tfrac{1}{2}, -1\right)$$

$$\left.\frac{df}{dt}\right|_{t=1} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\left.\frac{dy}{dt}\right|_{t=1}$$

$$= \nabla f(2,1) \cdot (x'(1), y'(1))$$

$$= \frac{1}{2} \cdot 3 + (-1)(-4) = \frac{11}{2}$$

$$\boxed{\text{Rate of change: } 5.5 \ \frac{\text{miles}}{\text{hour}}.}$$

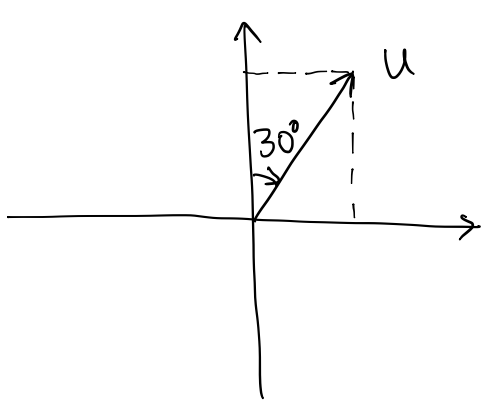
3 (b) Later Felix stops and takes a break at the point $(x, y) = (8, 2)$. Give the **unit** direction vector that points in the (x, y) direction Felix needs to initially walk in order to go steepest **downhill**.

$$\nabla f(8,2) = (2, -12) \qquad |\nabla f(8,2)| = \sqrt{4 + 144} = 2\sqrt{37}$$

The direction of fastest decrease is

$$-\frac{\nabla f}{|\nabla f|} = \frac{(-2, 12)}{4\sqrt{37}} = \boxed{\frac{(-1, 6)}{\sqrt{37}}}$$

3 (c) Felix is still standing at $(x, y) = (8, 2)$. He decides to walk in the direction that is 30 degrees east of north. Find the slope in this direction.



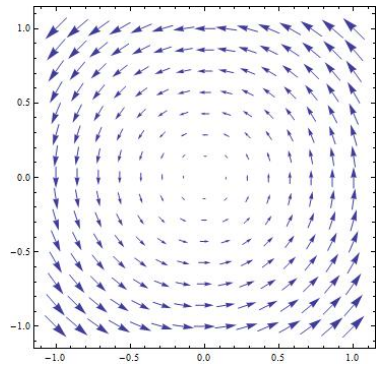
$$\vec{u} = (\sin 30°, \cos 30°) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

$$\text{Slope} = \nabla f(8,2) \cdot \vec{u} = (2, -12) \cdot \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = \boxed{1 - 6\sqrt{3} \ \ (\text{miles/hour})}$$

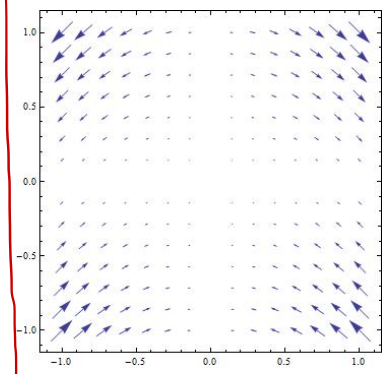
2. (10 points)

3 (a) Circle the picture that corresponds to the vector field $\mathbf{F}(x, y) = \langle xy, -x^2y \rangle$ on $\mathbb{R}^2$.

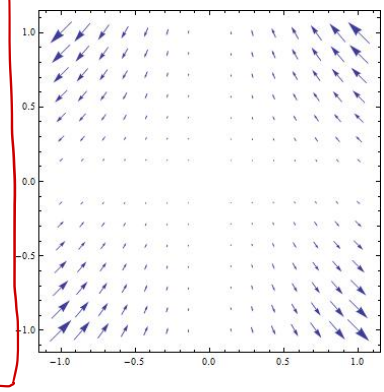
I



II



III



7 (b) Compute $\int_C \mathbf{H} \cdot d\mathbf{r}$ where $\mathbf{H}(x, y) = \left\langle x^2 + \frac{1}{y}, e^y \right\rangle$ and C is the portion of the circle $x^2 + y^2 = 1$ that lies in the first quadrant traversed counterclockwise.

C: $x = \cos t, \quad y = \sin t \quad (0 \leq t \leq \pi/2)$

$$\int_C \vec{H} \cdot d\vec{r} = \int_0^{\pi/2} \left(x^2 + \frac{1}{y}\right) x' + e^y y' \, dt$$

$$= \int_0^{\pi/2} \left(\cos^2 t + \frac{1}{\sin t}\right)(-\sin t) + (e^{\sin t})(\cos t) \, dt$$

$$= \frac{\cos^3 t}{3} - t + e^{\sin t} \Big|_{t=0}^{\pi/2}$$

$$= \left(0 - \frac{\pi}{2} + e^1\right) - \left(\frac{1}{3} - 0 + e^0\right)$$

$$= \boxed{-\frac{\pi}{2} + e - \frac{4}{3}}$$

6

3. (10 points) Let $\mathbf{F}(x, y, z) = \langle x^3 - 3xy^2, y^3 - 3x^2y, 30z\rangle$.

(a) Find a function $f(x, y, z)$ such that $\mathbf{F} = \nabla f$.

$$\frac{\partial f}{\partial x} = x^3 - 3xy^2 \Rightarrow f = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + C(y,z)$$

$$\Downarrow$$

$$\frac{\partial f}{\partial y} = y^3 - 3x^2y \qquad \frac{\partial f}{\partial y} = -3x^2y + \frac{\partial C}{\partial y}(y,z)$$

$$\Downarrow$$

$$\frac{\partial f}{\partial z} = 30z$$

$$\frac{\partial C}{\partial y} = y^3 \Rightarrow C(y,z) = \frac{y^4}{4} + C(z)$$

$$\Rightarrow \frac{\partial f}{\partial z} = 0 - 0 + 0 + \frac{dC}{dz}(z)$$

$$\Rightarrow \frac{dC}{dz} = 30z \Rightarrow C(z) = 15z^2 + C$$

Conclusion:

$$f = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + \frac{y^4}{4} + 15z^2$$

(can take $C=0$)

(b) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the portion of the helix

4

$$\begin{cases} x = 3\cos t \\ y = 3\sin t \\ z = t \end{cases}$$

with $0 \le t \le \pi$.

$$\vec{r}(\pi) = (-3, 0, \pi)$$
$$\vec{r}(0) = (3, 0, 0)$$

$$\int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(\pi)) - f(\vec{r}(0))$$

$$= \left(\cancel{\frac{(-3)^4}{4}} - 0 + 0 + 15\pi^2\right) - \left(\cancel{\frac{3^4}{4}} - 0 + 0 + 0\right)$$

$$= \boxed{15\pi^2}$$

4. (10 points) Let C be a simple closed curve going counterclockwise around a region R of $\mathbb{R}^2$. Let $M = M(x, y)$ be function defined on R.

5 (a) Express $\int_C M dx$ as a double integral over R.

$$\int_C M dx + 0 \cdot dy = \iint_R \left(0 - \frac{\partial M}{\partial y}\right) dA = \boxed{\iint_R \left(-\frac{\partial M}{\partial y}\right) dA}$$

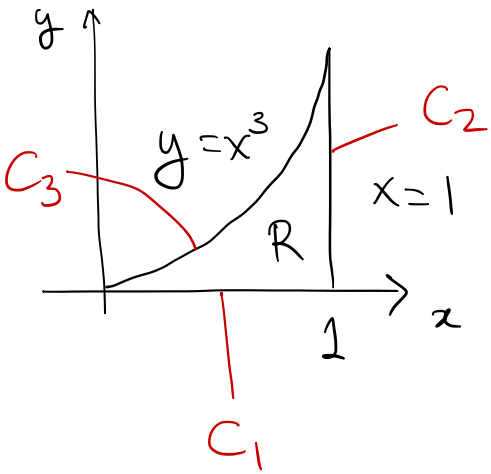
5 (b) Find M so that $\int_C M dx$ is the mass of R with densitiy $\rho(x, y) = (x + y)^2$.

<u>Want</u>: $-\frac{\partial M}{\partial y} = (x+y)^2$

$$\boxed{M = -\frac{(x+y)^3}{3}}$$ works.

5. (10 points) Consider the region R enclosed by the x-axis, $x = 1$ and $y = x^3$.

4 (a) Use the normal form of Green's theorem to find the flux of $\mathbf{F} = (1 + y^2)\hat{\mathbf{j}}$ out of R.



$$\operatorname{div}\vec{F} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(1+y^2) = 2y$$

$$\text{Flux} = \int_C \vec{F}\cdot\vec{N}\,ds = \iint_R \operatorname{div}(\vec{F})\,dA$$

$$= \iint_R 2y\,dA = \int_{x=0}^{1}\int_{y=0}^{x^3} 2y\,dy\,dx$$

$$= \int_0^1 \left[y^2\right]_0^{x^3} dx = \frac{x^7}{7}\Big|_0^1 = \boxed{\frac{1}{7}}$$

3 (b) Find the flux out of R through the two sides C_1 (the horizontal segment) and C_2 (the vertical segment).

On C_1: $\vec{N}$ = Outward unit normal $= (0,-1)$

On C_2: $\vec{N} = (1,0)$

$$\int_{C_1}\vec{F}\cdot\vec{N}\,ds = \int_{x=0}^{1}(0,1+0^2)\cdot(0,-1)\,dx = 1\cdot(-1) = \boxed{-1}$$

$$\int_{C_2}\vec{F}\cdot\vec{N}\,ds = \int_{y=0}^{1}(0,1+y^2)\cdot(1,0)\,dy = \boxed{0}$$

3 (c) Use parts (a) and (b) to find the flux out of the third side C_3.

$$\underbrace{\int_{C_1}\vec{F}\cdot\vec{N}\,ds}_{-1} + \underbrace{\int_{C_2}\vec{F}\cdot\vec{N}\,ds}_{0} + \int_{C_3}\vec{F}\cdot\vec{N}\,ds = \underbrace{\text{Flux out of } R}_{1/7}$$

$$\boxed{\int_{C_3}\vec{F}\cdot\vec{N}\,ds = \frac{8}{7}}$$